

SYDNEY GIRLS HIGH SCHOOL



2007 HSC Assessment Task 2

March 5, 2007

MATHEMATICS Extension 2

Year 12

Reading Time 5 minutes
Time allowed: 90 minutes

Topics: Complex Numbers

DIRECTIONS TO CANDIDATES:

- Attempt all questions
- Questions are not of equal value
- There are 3 questions with part marks shown
- All necessary working must be shown. Marks may be deducted for careless or badly arranged work
- Board approved calculators may be used
- Write on one side of the paper only

QUESTION 1 (Start a new page) 24 marks

a) Solve $2x^2 - x + 3 = 0$ and express your answer in the form $x + iy$ [2]

b) If $z_1 = 2 + 3i$ and $z_2 = 3 - 5i$, find in $x+iy$ form

i) $\overline{z_2}$ [1]

ii) $|z_1|$ [1]

iii) $3z_1 - 2z_2$ [2]

iv) $\frac{z_1}{z_2}$ [2]

c) Find $\sqrt{25 - 24i}$ and express your answer in $x+iy$ form. [3]

d) If $z = -3 + i$

i) Express z in modulus-argument form [2]

ii) Evaluate z^6 and express your answer in $x+iy$ form [2]

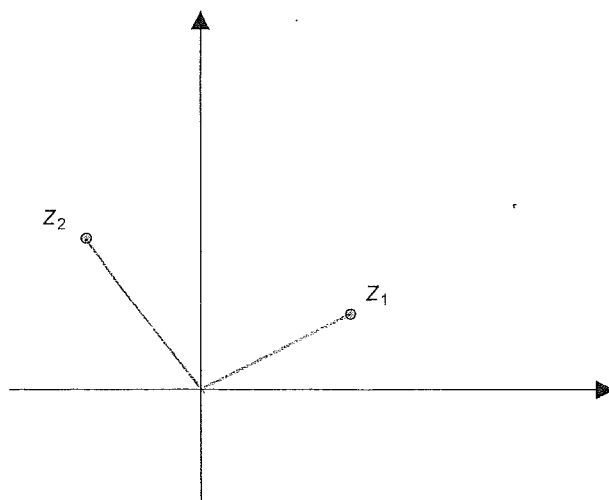
e) If $z_1 = a + ib$ and $z_2 = c + id$, prove that $|z_1 z_2| = |z_1| |z_2|$ [3]

f) If $z = r \operatorname{cis} \theta$, what is the modulus and argument of $rkiz$ where k is a non zero real number. [2]

g) Solve for x if $ix^2 - (1 + i)x + 2i - 3 = 0$ and express your solutions in $x+iy$ form. [4]

QUESTION 2 (Start a new page) 24 marks

a) Copy (or trace) the diagram below



On your diagram mark the positions of

- i) $z_1 + z_2$ [1]
 - ii) $z_2 - z_1$ [2]
 - iii) $-iz_2$ [2]
 - iv) $(1+i)z_2$ [2]
- b) i) Use a diagram to show that for any complex numbers z and w that $|z + w| \leq |z| + |w|$ [2]
- ii) Give a numerical example so that $|z + w| = |z| + |w|$ [1]
- c) Prove, by induction that $(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$ for positive n . [4]
- d) i) Solve $z^5 + 1 = 0$ over the complex field [2]
- ii) Sketch the solutions to $z^5 + 1 = 0$ on an Argand Diagram [1]
- iii) Let α be the root with the smallest positive argument and write all roots in terms of α [1]
- iv) Factorise $z^5 + 1$ over
- α) The complex field [1]
 - β) The real field [2]
 - γ) the rational field [1]
- v) Find the perimeter of the polygon whose vertices are represented by the roots of $z^5 + 1 = 0$ on the Argand Diagram. [2]

QUESTION 3 (Start a new page) 26 marks

a) If $z = x + iy$ express $\frac{z+i}{z-i}$ in the form $a + ib$ [3]

b) i) If $z = \cos \theta + i \sin \theta$, show that $z^n + \frac{1}{z^n} = 2 \cos n\theta$ [2]

ii) Express $\cos^4 \theta$ in terms of cosines of multiples of θ [3]

iii) Find $\int \cos^4 \theta . d\theta$ {given $\int \cos \theta . d\theta = \sin \theta + c$ } [2]

c) If w is a complex root of $z^3 - 1 = 0$,

i) Show that $1 + w + w^2 = 0$ [2]

ii) Find a quadratic equation whose roots are $1 + w$ and $1 + w^2$ [2]

iii) Evaluate $(1 + w)(1 + w^2)(1 + w^3)(1 + w^4)(1 + w^5)$ [2]

d) Express $-1 + \sqrt{3}i$ and $-1 - \sqrt{3}i$ in modulus-argument form, and hence prove that if n is a multiple of 3 then $(-1 + \sqrt{3}i)^n + (-1 - \sqrt{3}i)^n = 2^{n+1}$ [3]

e) Find all the solutions of $z^2 = \bar{iz}$ where z is a complex number [3]

f) Express $\cos 4\theta$ and $\sin 4\theta$ in terms of $\cos \theta$ and $\sin \theta$ and hence express $\tan 4\theta$ in terms of $\tan \theta$ [4]

-----end of paper-----

$$1(a) x = \frac{1 \pm \sqrt{1-24}}{4}$$

$$= \frac{1 \pm \sqrt{23}i}{4}$$

$$(1) i) 3+5i$$

$$ii) \sqrt{2^2 + 3^2}$$

$$= \sqrt{13}$$

$$iii) 6+9i - (6-10i)$$

$$= 19i$$

$$iv) \frac{2+3i}{3-5i} \times \frac{3+5i}{3+5i}$$

$$= \frac{6+10i+9i-15}{9+25}$$

$$= \frac{-9+19i}{34}$$

$$(c) (x+iy)^2 = 25-24i$$

$$x^2 - y^2 = 25 \quad 2xy = -24$$

$$y = -\frac{12}{x}$$

$$x^2 - \frac{144}{x^2} = 25$$

$$x^4 - 25x^2 - 144 = 0$$

$$x^2 = \frac{25 \pm \sqrt{625 + 576}}{2}$$

$$= \frac{25 \pm \sqrt{1201}}{2}$$

$$x^2 = \frac{25 + \sqrt{1201}}{2}$$

$$x = \pm \sqrt{\frac{25 + \sqrt{1201}}{2}} \quad y = \mp 12 \sqrt{\frac{2}{25 + \sqrt{1201}}}$$

$$\therefore \sqrt{25-24i} = \pm \left(\sqrt{\frac{25 + \sqrt{1201}}{2}} - i 12 \sqrt{\frac{2}{25 + \sqrt{1201}}} \right)$$

$$= \pm (5.46... - 2.19...i)$$

$$\frac{1}{5} (d) i) |z| = \sqrt{3^2 + 1^2}$$

$$= \sqrt{10}$$

$$\tan(\arg z) = \frac{1}{-3}$$

$$\therefore \arg z = 161.565^\circ$$

$$\approx 162^\circ$$

$$ii) z^6 = (\sqrt{10})^6 \operatorname{cis}(6 \times 162^\circ)$$

$$= 1000 \operatorname{cis}(972^\circ)$$

$$x = 1000 \cos(252^\circ)$$

$$= -309.0169944$$

$$y = 1000 \sin(252^\circ)$$

$$= -951.0565163$$

$$\therefore z^6 = -309 - 951i$$

$$(e) |z_1 z_2|$$

$$= |ac + iad + ibc - bd|$$

$$= \sqrt{(ac-bd)^2 + (ad+bc)^2}$$

$$= \sqrt{a^2c^2 + b^2d^2 + a^2d^2 + b^2c^2}$$

$$|z_1| |z_2|$$

$$= \sqrt{a^2 + b^2} \cdot \sqrt{c^2 + d^2}$$

$$= \sqrt{a^2c^2 + a^2d^2 + b^2c^2 + b^2d^2}$$

$$= |z_1 z_2| \quad \text{Q.E.D.}$$

$$(f) rki, r \operatorname{cis} \theta$$

$$= r^2 k \operatorname{cis}(\theta + \frac{\pi}{2})$$

$$\therefore |rki z| = r^2 k$$

$$\arg(rki z) = \theta + \frac{\pi}{2}$$

$$(g) x = \frac{(1+i)^{\frac{1}{2}} \sqrt{(1+i)^2 - 4 \times i(2i-3)}}{2i}$$

$$= \frac{(1+i)^{\frac{1}{2}} \sqrt{1+2i-1+8+12i}}{2i}$$

$$= \frac{(1+i)^{\frac{1}{2}} \sqrt{9+14i}}{2i}$$

$$\text{Let } 9+14i = (a+ib)^2$$

$$a^2 - b^2 = 9 \quad 2ab = 14$$

$$b = \frac{7}{a}$$

$$a^2 - \frac{49}{a^2} = 9$$

$$a^4 - 9a^2 - 49 = 0$$

$$a^2 = \frac{9 \pm \sqrt{64+196}}{2}$$

$$= \frac{9 \pm \sqrt{260}}{2}$$

$$= 4 \pm \sqrt{65}$$

$$a = \pm \sqrt{4 + \sqrt{65}}$$

$$= \pm 3.4730...$$

$$b = \frac{7}{a}$$

$$= \pm \frac{7}{3.4730...}$$

$$= \pm 2.0155...$$

$$\therefore x = \frac{1 \pm \sqrt{4 + \sqrt{65}}}{2}$$

$$+ \frac{i \pm \frac{7}{\sqrt{4 + \sqrt{65}}}}{2i}$$

$$= \frac{1 \pm \sqrt{4 + \sqrt{65}}}{2}$$

$$+ \frac{i \pm \frac{7}{\sqrt{4 + \sqrt{65}}}}{2} i$$

$$= 1.57... - 2.23...i$$

or

$$-0.50... + 1.23...i$$

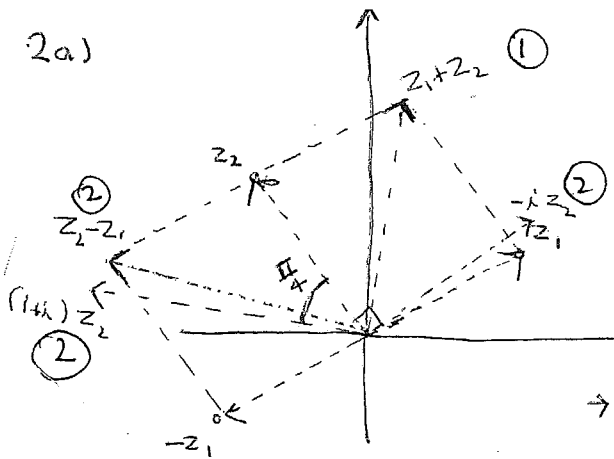
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EXT 2

TASK 2

2007

2a)



$$\frac{2}{5} \text{ d) i) } z^5 = -1$$

$$r^5 \cos 5\theta = -1$$

$$r = \sqrt{1^2 + 0^2} \quad \cos 5\theta = -1 \quad \sin 5\theta = 0$$

$$= 1$$

$$\therefore 5\theta = \pi, 3\pi, 5\pi, 7\pi, 9\pi$$

$$\theta = \frac{\pi}{5}, \frac{3\pi}{5}, \pi, \frac{7\pi}{5}, \frac{9\pi}{5}$$

$$z = \cos \frac{\pi}{5}, \cos \frac{3\pi}{5}, \cos \pi, \cos \frac{7\pi}{5}, \cos \frac{9\pi}{5} \quad (2)$$

$$\text{iii) } \alpha = \cos \frac{\pi}{5} \quad \cos \frac{3\pi}{5} = \alpha^3 \quad \cos \pi = \alpha^5 \quad \cos \frac{7\pi}{5} = \alpha^7 = \alpha^2$$

$$\cos \frac{9\pi}{5} = \alpha^9 = -\alpha^4 \quad (1)$$

$$\text{iv) } \alpha) (z+1)(z - \cos \frac{\pi}{5} - i \sin \frac{\pi}{5})(z - \cos \frac{\pi}{5} + i \sin \frac{\pi}{5})$$

$$(z - \cos \frac{3\pi}{5} - i \sin \frac{3\pi}{5})(z - \cos \frac{3\pi}{5} + i \sin \frac{3\pi}{5}) \quad (1)$$

$$\beta) (z+1)(z^2 - 2z \cos \frac{\pi}{5} + \cos^2 \frac{\pi}{5} + \sin^2 \frac{\pi}{5})$$

$$(z - 2z \cos \frac{3\pi}{5} + \sin^2 \frac{3\pi}{5} + \cos^2 \frac{3\pi}{5}) \quad (2)$$

$$= (z+1)(z^2 - 2z \cos \frac{\pi}{5} + 1)(z^2 - 2z \cos \frac{3\pi}{5} + 1)$$

v)

$$\begin{array}{r} z^5 + 1 \\ z+1 \overline{) z^5} \\ \underline{z^5 + z^4} \\ -z^4 \\ \underline{-z^4 - z^3} \\ z^3 \\ \underline{z^3 + z^2} \\ -z^2 \\ \underline{-z^2 - z} \\ z+1 \\ \underline{z+1} \\ 0 \end{array}$$

$$\therefore z^5 + 1 = (z+1)(z^4 - z^3 + z^2 - z + 1) \quad (1)$$

vi) let s be a side of the regular pentagon

$$s^2 = 1^2 + 1^2 - 2 \times 1 \times 1 \times \cos \frac{2\pi}{5}$$

$$= 2 - 2 \cos \frac{2\pi}{5}$$

$$s = \sqrt{2 - 2 \cos \frac{2\pi}{5}}$$

$$P = 5 \sqrt{2 - 2 \cos \frac{2\pi}{5}} \quad (2)$$

$$\approx 5.88$$

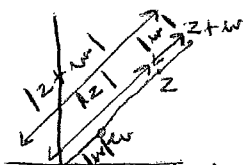
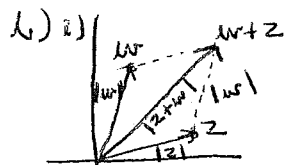
(4)

$$\text{iv) } |1+i| = \sqrt{1^2 + 1^2}$$

$$= \sqrt{2}$$

$$\tan [\arg(1+i)] = \frac{1}{1}$$

$$\therefore \arg(1+i) = \frac{\pi}{4}$$



Since one side of a triangle is less than the sum of the other two sides.

$$\therefore |z+w| \leq |z| + |w|$$

$$\text{ii) let } z = w = 0$$

$$|0+0| = |0| + |0| \quad (1)$$

c) 1. Prove for $n=1$

$$(\cos \theta + i \sin \theta)^1 = \cos \theta + i \sin \theta$$

$$\cos \theta + i \sin \theta = \cos \theta + i \sin \theta$$

$\therefore$ true for $n=1$

2. Assume true for $n=k$

$$\therefore (\cos \theta + i \sin \theta)^k = \cos k\theta + i \sin k\theta$$

3. Prove for $n=k+1$

$$(\cos \theta + i \sin \theta)^{k+1}$$

$$= (\cos \theta + i \sin \theta)^k \cdot (\cos \theta + i \sin \theta)$$

$$= (\cos k\theta + i \sin k\theta)(\cos \theta + i \sin \theta)$$

$$= \cos k\theta \cos \theta + i \cos k\theta \sin \theta + i \sin k\theta \cos \theta + i^2 \sin k\theta \sin \theta$$

$$= \cos k\theta \cos \theta - \sin k\theta \sin \theta + i (\sin k\theta \cos \theta + \cos k\theta \sin \theta)$$

$$= \cos[(k+1)\theta] + i \sin[(k+1)\theta]$$

$\therefore$ true for $n=k+1$

4. Since true for $n=1$ and $n=k+1$

$\therefore$ true for $n \geq 1$

Question 3.

3/5

a) $z = \frac{x+iy+i}{x+iy-i}$

conjugate of
denom is (in terms of z)
 $(\bar{z} + i)$

$$= \frac{x+i(y+1)}{x+i(y-1)} \times \frac{x-i(y-1)}{x-i(y-1)} \quad \checkmark$$

$$= \frac{x^2 - i(y-1)x + i(y+1)x - i^2(y+1)(y-1)}{x^2 - i^2(y-1)^2}$$

$$= \frac{x^2 - iyx + ix + iyx + i^2x + y^2 - 1}{x^2 - i^2(y-1)^2} \quad \checkmark$$

$$= \frac{x^2 + y^2 - 2y + 1}{x^2 + y^2 - 2y + 1} \left[\frac{x^2 + y^2 - 1}{x^2 + y^2 - 2y + 1} + \frac{2ix}{x^2 + y^2 - 2y + 1} \right] \quad (3)$$

b) i) $z = \cos \theta + i \sin \theta$

$$z^n = \cos(n\theta) + i \sin(n\theta) \quad (1) \quad z^{-n} = \cos(-n\theta) + i \sin(-n\theta)$$

$$z^{-n} = \cos(n\theta) - i \sin(n\theta) \quad (2) \quad (\cos \theta \text{ even } \sin \theta \text{ odd})$$

$$(1)+(2) \quad z^n + \frac{1}{z^n} = 2 \cos(n\theta)$$

ii) put $n=1$ in above

$$z + \frac{1}{z} = 2 \cos(\theta)$$

$$\left(z + \frac{1}{z}\right)^4 = 2^4 \cos^4 \theta \quad \checkmark$$

$$16 \cos^4 \theta = z^4 + 4z^2 + 6 + \frac{4}{z^2} + \frac{1}{z^4} \quad \checkmark$$

$$\cos^4 \theta = \frac{1}{16} \left(z^4 + \frac{1}{z^4} + 4\left(z^2 + \frac{1}{z^2}\right) + 6 \right) \quad (3)$$

$$\text{put } n=4, 2$$

$$\cos^4 \theta = \frac{1}{16} (2 \cos 4\theta + 8 \cos 2\theta + 6) \quad \checkmark$$

$$\text{iii) } \int \cos^4 \theta d\theta = \frac{1}{16} \int 2 \cos 4\theta + 8 \cos 2\theta + 6 d\theta$$

$$= \frac{1}{16} \left(\frac{1}{2} \sin 4\theta + 4 \sin 2\theta + 6\theta \right) + C \quad (2) \quad \checkmark \checkmark$$

c) i) $1, \omega, \omega^2$ are roots of $z^3 - 1$ and sum of roots is given by $-\frac{b}{a} = \frac{0}{1} = 0$ $\checkmark \quad (2)$

OR any other soln

ii) Sum of roots $= (1 + \omega + \omega^2) + 1$

$$= 1 \quad \checkmark \quad (2)$$

Product of roots $= (1 + \omega)(1 + \omega^2)$ $12x^2 - x + 1 = 0$

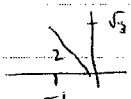
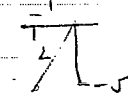
$$= 1 + \omega + \omega^2 + \omega^3$$

$$= 1$$

$$\begin{aligned}
 c) \text{ ii) } & (1+w)(1+w^2)(1+w^3)(1+w^4)(1+w^5) \\
 &= (1+w)(1+w^2)(2)(1+w)(1+w^4) \\
 &= 2(1+w+w^2+w^3)(1+w+w^2+w^3) \\
 &= 2(1)(1) \\
 &= 2
 \end{aligned}$$

[Note $2w^3$
award 2 marks]

Quick Soln
 $= (-w^2)(-w)2(-w^2)(-1)$
 $= w^3 \times 2 \times w^3 = 2$

d) $-1 + \sqrt{3}i$  , $-1 - \sqrt{3}i$ 

$$= 2 \operatorname{cis} \frac{2\pi}{3}$$

$$= 2 \operatorname{cis} \left(-\frac{2\pi}{3}\right)$$

$$\begin{aligned}
 \text{then } (-1 + \sqrt{3}i)^n + (-1 - \sqrt{3}i)^n &= \left(2 \operatorname{cis} \frac{2\pi}{3}\right)^n + \left(2 \operatorname{cis} -\frac{2\pi}{3}\right)^n \\
 &= 2^n \operatorname{cis} \frac{2\pi n}{3} + 2^n \operatorname{cis} -\frac{2\pi n}{3}
 \end{aligned}$$

Note: proving true
for $n=3$ (only)

is awarded 2 (max)
marks only

$$\begin{aligned}
 &= 2^n (\cos \frac{2\pi n}{3} + i \sin \frac{2\pi n}{3} + \cos -\frac{2\pi n}{3} + i \sin -\frac{2\pi n}{3}) \\
 &= 2^n (1 + 0 + 1 + 0) \\
 &= 2^n (2) \\
 &= 2^{n+1}
 \end{aligned}$$

since n a
multiple of 3

e) $z^2 = \bar{z}$

$$(x+iy)^2 = \overline{i(x+iy)}$$

$$x^2 - y^2 + 2ixy = ix - y$$

$$x^2 - y^2 + 2ixy = -y - ix$$

$$x^2 - y^2 + y + i(2yx + x) = 0$$

$$x^2 - y^2 + y = 0 \quad 2yx + x = 0 \quad \text{equate Re, Im}$$

$$x(2y+1) = 0$$

$$x = 0, y = -\frac{1}{2}$$

when $x=0$

$$-y^2 + y = 0$$

$$y(1-y) = 0 \quad y=0, y=1 \quad (3)$$

when $y = -\frac{1}{2}$

$$x^2 - \frac{1}{4} - \frac{1}{2} = 0$$

$$\begin{aligned}
 x^2 &= \frac{3}{4} \\
 x &= \pm \frac{\sqrt{3}}{2}
 \end{aligned}$$

ie
or

$$x=0, y=0$$

$$x=0, y=1$$

$$x = \frac{\sqrt{3}}{2}, y = -\frac{1}{2}$$

$$x = -\frac{\sqrt{3}}{2}, y = -\frac{1}{2}$$

1 mark

1 mark

$$f) (\cos \theta + i \sin \theta)^4 = (\cos 4\theta + i \sin 4\theta) \text{ [De Moivre's Th]} \\ = \cos^4 \theta + 4 \cos^3 \theta i \sin \theta + 6 \cos^2 \theta i^2 \sin^2 \theta + 4 \cos \theta i^3 \sin^3 \theta + i^4 \sin^4 \theta \\ = \cos^4 \theta + 4 \cos^3 \theta i \sin \theta - 6 \cos^2 \theta \sin^2 \theta - 4 \cos \theta i \sin^3 \theta + \sin^4 \theta \quad \checkmark$$

Equate Re

$$\cos 4\theta = \cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta \quad \checkmark$$

Equate Im

$$\sin 4\theta = 4 \cos^3 \theta \sin \theta - 4 \cos \theta \sin^3 \theta \quad \checkmark$$

$$\tan 4\theta = \frac{\sin 4\theta}{\cos 4\theta}$$

④

$$= \frac{4 \cos^3 \theta \sin \theta - 4 \cos \theta \sin^3 \theta}{\cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta} \quad \begin{array}{l} \div \cos^4 \theta \\ \div \cos^4 \theta \end{array}$$

$$= \frac{4 \tan \theta - 4 \tan^3 \theta}{1 - 6 \tan^2 \theta + \tan^4 \theta} \quad \checkmark$$